OPERATING SYSTEMS – **CS252M**

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Project Demo

Presented by:

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PROBLEM STATEMENT:

The **USBGuard Kernel Module** is a Linux kernel module that provides a robust mechanism to monitor and restrict USB device connections. It allows dynamic configuration of rules to authorize or block USB devices based on their Vendor ID (VID), Product ID (PID), class, or serial number. The module integrates with sysfs for user interaction and logs all connection and disconnection events.

This addresses critical security concerns related to:

- Unauthorized USB device connections
- Potential malware delivery via USB devices
- Data exfiltration through removable storage
- BadUSB attacks and similar threats

OBJECTIVE:

The primary objectives of this project were:

- To implement a kernel module capable of enforcing USB device access policies in real time.
- To provide dynamic rule management through sysfs and rule files.
- To log connection and disconnection events for security auditing.
- To restrict unauthorized USB devices based on hardware identifiers.
- To maintain lightweight operation with minimal kernel overhead.

FILE STRUCTURE:

- **usbguard.c**: Main source file containing the kernel module implementation.
- **Makefile**: Build script for compiling and managing the kernel module.
- **usbguard.rules**: Default rule file with sample configurations.
- **install.sh**: Installation script to set up the environment and copy necessary files.
- **README.md**: Documentation for the project.

MAKE FILE

A Makefile is used to automate the process of compiling, building, and cleaning projects. In this case, it helps in building and cleaning the Linux kernel module easily.

- **obj-m := usbguard.o**
  Specifies the kernel object (module) to be built. Here, the file usbguard.c will be compiled into a loadable kernel object usbguard.ko.
- **KDIR := /lib/modules/$(shell uname -r)/build**
  Defines the path to the kernel build directory of the currently running kernel. This ensures that the module is compiled for the correct kernel version.
- **PWD := $(shell pwd)**
  Stores the current working directory path where the source code of the module is present.
- **all:**
  This is the default target which triggers the build process of the kernel module using the kernel's build system.
- **clean:**
  Used to remove the temporary files generated during the build process such as .o, .ko, and .mod.c files. This ensures a clean environment for rebuilding the module.

WORKING:

1.Rule Management

- Reads allowed devices from /etc/usbguard.rules.
- Each rule consists of a **VID** and **PID** pair.
- Supports comments and blank lines for better readability and allows runtime rule addition through the sysfs interface.

2. USB Device Validation

- On each USB device connection (probe event), the module reads the device's VID/PID and optional serial number.
- It checks if the device exists in the allowed rules.
- If the device is not allowed, the connection is rejected by returning an error (-EACCES).

DEFINING OF RULES TO BE FOLLOWED(USBGUARD.RULES):

```
# USBGuard Rules File
# --------------------
# This file contains a list of allowed USB devices based on their Vendor ID (VID) and Product ID (PID).
# Each entry should be in the format: VID PID
# VID = Vendor ID (4-digit hexadecimal number)
# PID = Product ID (4-digit hexadecimal number)
#
# Lines starting with '#' are comments and will be ignored.
#
# Example entries:
# Allow a standard keyboard (example VID/PID)
04d9 1702
# Allow a specific USB flash drive (example VID/PID)
0781 5567
# Allow a Logitech mouse (example VID/PID)
046d c077
# more devices following the same format VID PID
```

VERIFYING RULES:

Snippet from usbguard.c

```
static bool match_rules(struct usb_device *udev)
{
    int i;
    u16 vid = le16_to_cpu(udev->descriptor.idVendor);
    u16 pid = le16_to_cpu(udev->descriptor.idProduct);

    mutex_lock(&rules_lock);
    for (i = 0; i < rule_count; i++) {
        if (rules[i].vid == vid && rules[i].pid == pid) {
            mutex_unlock(&rules_lock);
            return true;
        }
    }
    mutex_unlock(&rules_lock);
    return false;
}
```

WORKING:

3.Dynamic Sysfs Interface

- Two sysfs files are created under /sys/kernel/usbguard/:
  /sys/kernel/usbguard/rules — Lists or adds allowed VID/PID pairs.
  /sys/kernel/usbguard/blocked_serials — Lists or adds blocked serial numbers.

- This enables administrators to modify security rules on the fly without unloading the module.

4.Logging and Monitoring

- Connection and disconnection events are logged via printk() for auditing.

- Logs can be viewed using dmesg | grep usbguard.

ANALYSIS OF PROPOSED SOLUTION:

Security Benefits:

Real-Time Protection: Prevents unauthorized devices before they can be mounted or accessed by user-space.

Policy Enforcement: Rules enforce device-level access control similar to enterprise-grade USB management solutions.

Logging and Auditing: Provides visibility into device events for administrators.

ANALYSIS OF PROPOSED SOLUTION:

Performance Considerations
Lightweight Design: The module adds minimal overhead to the kernel.

Efficient Rule Matching: Simple array-based lookups ensure low latency.

Thread-Safe Access: Use of mutex_lock() ensures synchronization for concurrent rule modifications.

INSTALLING OF REQUIRED MODULES:

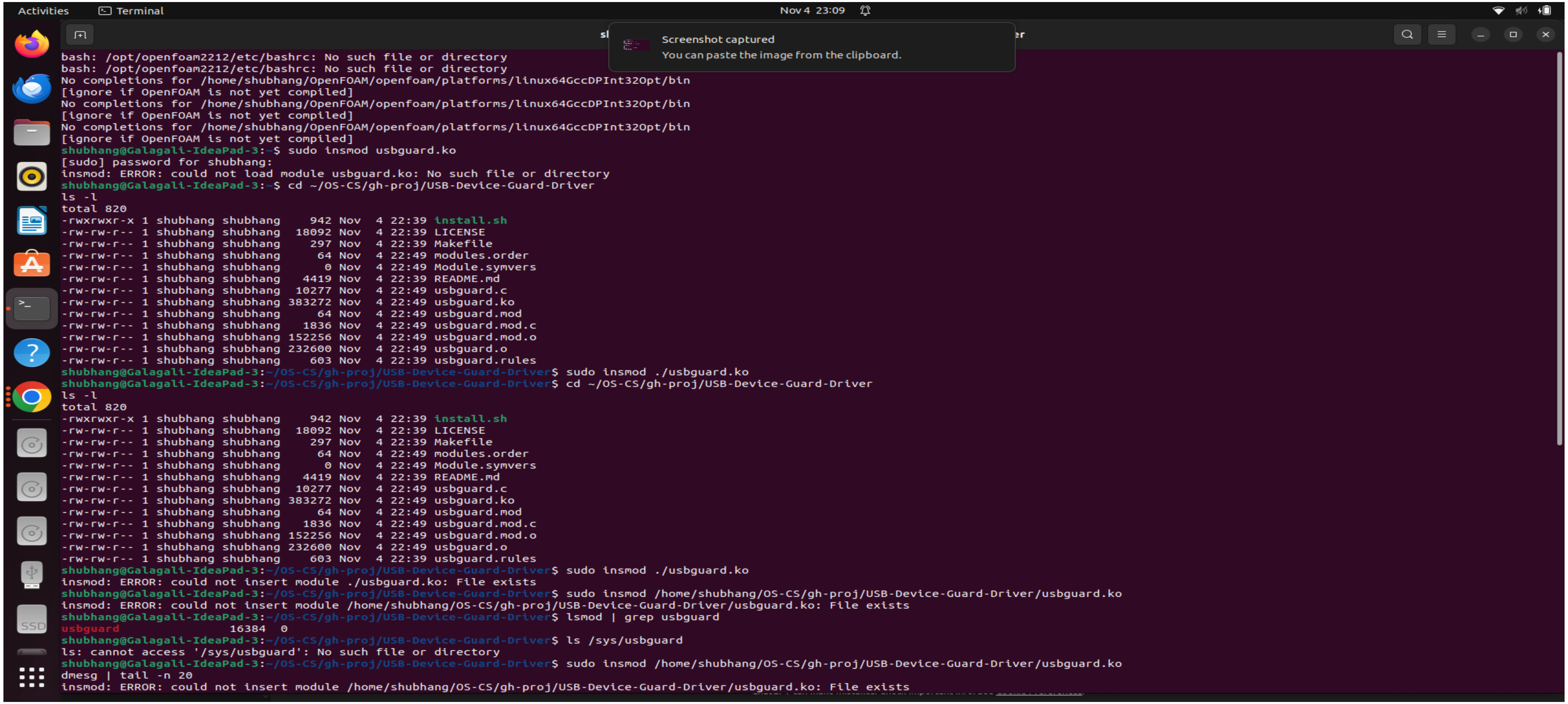
```
#!/bin/bash
set -e
# Define installation paths
MODULE_NAME="usbguard"
RULES_FILE="/etc/usbguard.rules"
KERNEL_MODULE_DIR="/lib/modules/$(uname -r)/extra"
# Ensure script is run as root
if [[ $EUID -ne 0 ]]; then
    echo "This script must be run as root." >&2
    exit 1
fi
echo "Installing USBGuard kernel module..."
# Copy predefined rules file if it does not already exist
if [[ ! -f "$RULES_FILE" ]]; then
    echo "Copying predefined usbguard rules file..."
    cp usbguard.rules "$RULES_FILE"
fi
```

INSTALLING OF REQUIRED MODULES:

```
# Build the kernel module
make clean
make

# Create module directory if it does not exist
mkdir -p "$KERNEL_MODULE_DIR"
# Copy module to kernel directory
cp "$MODULE_NAME.ko" "$KERNEL_MODULE_DIR/"
# Refresh module dependencies
depmod
# Load the module
modprobe "$MODULE_NAME"
# Ensure module loads at boot
echo "$MODULE_NAME" | tee -a /etc/modules > /dev/null
# Done
echo "USBGuard module installed and configured successfully."
```

EXECUTION:


Activities
Terminal
Nov 4 23:09
Screenshot captured
You can paste the image from the clipboard.
bash: /opt/openfoam2212/etc/bashrc: No such file or directory
bash: /opt/openfoam2212/etc/bashrc: No such file or directory
No completions for /home/shubhang/OpenFOAM/openfoam/platforms/linux64GccDPInt32Opt/bin
[ignore if OpenFOAM is not yet compiled]
No completions for /home/shubhang/OpenFOAM/openfoam/platforms/linux64GccDPInt32Opt/bin
[ignore if OpenFOAM is not yet compiled]
No completions for /home/shubhang/OpenFOAM/openfoam/platforms/linux64GccDPInt32Opt/bin
[ignore if OpenFOAM is not yet compiled]
shubhang@Galagali-IdeaPad-3:~$ sudo insmod usbguard.ko
[sudo] password for shubhang:
insmod: ERROR: could not load module usbguard.ko: No such file or directory
shubhang@Galagali-IdeaPad-3:~$ cd ~/OS-CS/gh-proj/USB-Device-Guard-Driver
ls -l
total 820
-rwxrwxr-x 1 shubhang shubhang 942 Nov 4 22:39 install.sh
-rw-rw-r-- 1 shubhang shubhang 18092 Nov 4 22:39 LICENSE
-rw-rw-r-- 1 shubhang shubhang 297 Nov 4 22:39 Makefile
-rw-rw-r-- 1 shubhang shubhang 64 Nov 4 22:49 modules.order
-rw-rw-r-- 1 shubhang shubhang 0 Nov 4 22:49 Module.symvers
-rw-rw-r-- 1 shubhang shubhang 4419 Nov 4 22:39 README.md
-rw-rw-r-- 1 shubhang shubhang 10277 Nov 4 22:49 usbguard.c
-rw-rw-r-- 1 shubhang shubhang 383272 Nov 4 22:49 usbguard.ko
-rw-rw-r-- 1 shubhang shubhang 64 Nov 4 22:49 usbguard.mod
-rw-rw-r-- 1 shubhang shubhang 1836 Nov 4 22:49 usbguard.mod.c
-rw-rw-r-- 1 shubhang shubhang 152256 Nov 4 22:49 usbguard.mod.o
-rw-rw-r-- 1 shubhang shubhang 232600 Nov 4 22:49 usbguard.o
-rw-rw-r-- 1 shubhang shubhang 603 Nov 4 22:39 usbguard.rules
shubhang@Galagali-IdeaPad-3:~/OS-CS/gh-proj/USB-Device-Guard-Driver$ sudo insmod ./usbguard.ko
shubhang@Galagali-IdeaPad-3:~/OS-CS/gh-proj/USB-Device-Guard-Driver$ cd ~/OS-CS/gh-proj/USB-Device-Guard-Driver
ls -l
total 820
-rwxrwxr-x 1 shubhang shubhang 942 Nov 4 22:39 install.sh
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-rw-rw-r-- 1 shubhang shubhang 297 Nov 4 22:39 Makefile
-rw-rw-r-- 1 shubhang shubhang 64 Nov 4 22:49 modules.order
-rw-rw-r-- 1 shubhang shubhang 0 Nov 4 22:49 Module.symvers
-rw-rw-r-- 1 shubhang shubhang 4419 Nov 4 22:39 README.md
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-rw-rw-r-- 1 shubhang shubhang 383272 Nov 4 22:49 usbguard.ko
-rw-rw-r-- 1 shubhang shubhang 64 Nov 4 22:49 usbguard.mod
-rw-rw-r-- 1 shubhang shubhang 1836 Nov 4 22:49 usbguard.mod.c
-rw-rw-r-- 1 shubhang shubhang 152256 Nov 4 22:49 usbguard.mod.o
-rw-rw-r-- 1 shubhang shubhang 232600 Nov 4 22:49 usbguard.o
-rw-rw-r-- 1 shubhang shubhang 603 Nov 4 22:39 usbguard.rules
shubhang@Galagali-IdeaPad-3:~/OS-CS/gh-proj/USB-Device-Guard-Driver$ sudo insmod ./usbguard.ko
insmod: ERROR: could not insert module ./usbguard.ko: File exists
shubhang@Galagali-IdeaPad-3:~/OS-CS/gh-proj/USB-Device-Guard-Driver$ sudo insmod /home/shubhang/OS-CS/gh-proj/USB-Device-Guard-Driver/usbguard.ko
insmod: ERROR: could not insert module /home/shubhang/OS-CS/gh-proj/USB-Device-Guard-Driver/usbguard.ko: File exists
shubhang@Galagali-IdeaPad-3:~/OS-CS/gh-proj/USB-Device-Guard-Driver$ lsmod | grep usbguard
usbguard 16384 0
shubhang@Galagali-IdeaPad-3:~/OS-CS/gh-proj/USB-Device-Guard-Driver$ ls /sys/usbguard
ls: cannot access '/sys/usbguard': No such file or directory
shubhang@Galagali-IdeaPad-3:~/OS-CS/gh-proj/USB-Device-Guard-Driver$ sudo insmod /home/shubhang/OS-CS/gh-proj/USB-Device-Guard-Driver/usbguard.ko
dmesg | tail -n 20
insmod: ERROR: could not insert module /home/shubhang/OS-CS/gh-proj/USB-Device-Guard-Driver/usbguard.ko: File exists

LOADING OF FILES:

```
bash: /opt/openfoam2212/etc/bashrc: No such file or directory
bash: /opt/openfoam2212/etc/bashrc: No such file or directory
No completions for /home/shubhang/OpenFOAM/openfoam/platforms/linux64GccDPInt32Opt/bin
[ignore if OpenFOAM is not yet compiled]
No completions for /home/shubhang/OpenFOAM/openfoam/platforms/linux64GccDPInt32Opt/bin
[ignore if OpenFOAM is not yet compiled]
No completions for /home/shubhang/OpenFOAM/openfoam/platforms/linux64GccDPInt32Opt/bin
[ignore if OpenFOAM is not yet compiled]
shubhang@Galagali-IdeaPad-3:~$ ls
Desktop     log.lammps  Open      Pictures  Templates
Documents   MicroSim    OpenFOAM  Public    triangular_cylinder
Downloads   Music       OS-CS     snap      Videos
shubhang@Galagali-IdeaPad-3:~$ cd OS-CS/
shubhang@Galagali-IdeaPad-3:~/OS-CS$ mkdir ~/OS_Project_Demo
shubhang@Galagali-IdeaPad-3:~/OS-CS$ cd ~/OS_Project_Demo
shubhang@Galagali-IdeaPad-3:~/OS_Project_Demo$ git clone https://github.com/galavashubhang/OS-Project.git
Cloning into 'OS-Project'...
remote: Enumerating objects: 82, done.
remote: Counting objects: 100% (82/82), done.
remote: Compressing objects: 100% (42/42), done.
remote: Total 82 (delta 39), reused 82 (delta 39), pack-reused 0 (from 0)
Receiving objects: 100% (82/82), 49.42 KiB | 1.18 MiB/s, done.
Resolving deltas: 100% (39/39), done.
shubhang@Galagali-IdeaPad-3:~/OS_Project_Demo$ ls
OS-Project
shubhang@Galagali-IdeaPad-3:~/OS_Project_Demo$ cd OS-Project/
shubhang@Galagali-IdeaPad-3:~/OS_Project_Demo/OS-Project$ ls
install.sh  LICENSE  Makefile  README.md  usbguard.c  usbguard.rules
shubhang@Galagali-IdeaPad-3:~/OS_Project_Demo/OS-Project$ make
make -C /lib/modules/6.8.0-45-generic/build M=/home/shubhang/OS_Project_Demo/OS-Project modules
make[1]: Entering directory '/usr/src/linux-headers-6.8.0-45-generic'
warning: the compiler differs from the one used to build the kernel
  The kernel was built by: x86_64-linux-gnu-gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04) 12.3.0
  You are using:           gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04.2) 12.3.0
  CC [M]  /home/shubhang/OS_Project_Demo/OS-Project/usbguard.o
  MODPOST /home/shubhang/OS_Project_Demo/OS-Project/Module.symvers
  CC [M]  /home/shubhang/OS_Project_Demo/OS-Project/usbguard.mod.o
  LD [M]  /home/shubhang/OS_Project_Demo/OS-Project/usbguard.ko
  BTF [M] /home/shubhang/OS_Project_Demo/OS-Project/usbguard.ko
Skipping BTF generation for /home/shubhang/OS_Project_Demo/OS-Project/usbguard.ko due to unavailability of vmlinux
make[1]: Leaving directory '/usr/src/linux-headers-6.8.0-45-generic'
shubhang@Galagali-IdeaPad-3:~/OS_Project_Demo/OS-Project$ sudo insmod usbguard.ko
[sudo] password for shubhang:
```

RESTRICTING A DEVICE:

```
[ 1734.473326] usbguard: module unloaded
[ 1832.544577] usbcore: registered new interface driver usbguard_demo
[ 1832.544584] usbguard: module loaded
[ 1963.420964] usbcore: deregistering interface driver usbguard_demo
[ 1963.421018] usbguard: module unloaded
[ 2442.398974] usbcore: registered new interface driver usbguard_demo
[ 2442.398980] usbguard: module loaded
[ 2772.694511] usbcore: deregistering interface driver usbguard_demo
[ 2772.694558] usbguard: module unloaded
[ 2781.395631] usbcore: registered new interface driver usbguard_demo
[ 2781.395635] usbguard: module loaded
[ 3259.673287] usbcore: deregistering interface driver usbguard_demo
[ 3259.673317] usbguard: module unloaded
[ 3271.580641] usbguard: device VID=0951 PID=1643 attached
[ 3271.580653] usbguard: VID/PID not allowed, rejecting device
[ 3271.580658] usbguard_demo: probe of 1-1:1.0 failed with error -13
[ 3271.580707] usbcore: registered new interface driver usbguard_demo
[ 3271.580709] usbguard: module loaded
shubhang@Galagali-IdeaPad-3:~/OS_Project_Demo/OS-Project$ sudo rmmod usbguard_demo
rmmod: ERROR: Module usbguard_demo is not currently loaded
shubhang@Galagali-IdeaPad-3:~/OS_Project_Demo/OS-Project$ sudo rmmod usbguard
shubhang@Galagali-IdeaPad-3:~/OS_Project_Demo/OS-Project$ lsmod | grep -i usbguard
shubhang@Galagali-IdeaPad-3:~/OS_Project_Demo/OS-Project$ sudo insmod usbguard.ko
shubhang@Galagali-IdeaPad-3:~/OS_Project_Demo/OS-Project$ lsusb -t
/:  Bus 02.Port 1: Dev 1, Class=root_hub, Driver=xhci_hcd/6p, 10000M
/:  Bus 01.Port 1: Dev 1, Class=root_hub, Driver=xhci_hcd/12p, 480M
    |__ Port 1: Dev 18, If 0, Class=Mass Storage, Driver=usb-storage, 480M
    |__ Port 2: Dev 3, If 0, Class=Human Interface Device, Driver=usbhid, 1.5M
    |__ Port 2: Dev 3, If 1, Class=Human Interface Device, Driver=usbhid, 1.5M
    |__ Port 5: Dev 4, If 1, Class=Video, Driver=uvcvideo, 480M
    |__ Port 5: Dev 4, If 0, Class=Video, Driver=uvcvideo, 480M
    |__ Port 9: Dev 5, If 0, Class=Human Interface Device, Driver=usbhid, 1.5M
    |__ Port 10: Dev 6, If 0, Class=Wireless, Driver=btusb, 12M
    |__ Port 10: Dev 6, If 1, Class=Wireless, Driver=btusb, 12M
```

LOADING AND VERIFYING DETAILS OF DEVICE INSERTED:

```
ce= 0.00
[ 7428.893560] usb 1-9: New USB device strings: Mfr=1, Product=2, SerialNumber=3
[ 7428.893561] usb 1-9: Product: DataTraveler G3
[ 7428.893563] usb 1-9: Manufacturer: KINGSTON
[ 7428.893564] usb 1-9: SerialNumber: 001CC07CEDAEACA1691804D2
[ 7428.895473] usb-storage 1-9:1.0: USB Mass Storage device detected
[ 7428.896086] scsi host1: usb-storage 1-9:1.0
[ 7430.735033] scsi 1:0:0:0: Direct-Access     KINGSTON DataTraveler G3  1.00 PQ: 0
ANSI: 4
[ 7430.735555] sd 1:0:0:0: Attached scsi generic sg1 type 0
[ 7430.736780] sd 1:0:0:0: [sdb] 15204352 512-byte logical blocks: (7.78 GB/7.25 GiB
)
[ 7430.738797] sd 1:0:0:0: [sdb] Write Protect is off
[ 7430.738801] sd 1:0:0:0: [sdb] Mode Sense: 2f 00 00 00
[ 7430.740793] sd 1:0:0:0: [sdb] Write cache: disabled, read cache: enabled, doesn't
 support DPO or FUA
[ 7430.747252]  sdb: sdb1
[ 7430.747453] sd 1:0:0:0: [sdb] Attached SCSI removable disk
[ 7431.106610] FAT-fs (sdb1): Volume was not properly unmounted. Some data may be co
rrupt. Please run fsck.
[ 7482.312897] usbcore: deregistering interface driver usbguard_demo
[ 7482.312930] usbguard: Module Unloaded.
[ 7482.339553] usbcore: registered new interface driver usbguard_demo
[ 7482.339559] usbguard: Kernel Module Registered and Active.
```

Thank You!

Submitted to:

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